Relative Performance Feedback and Redistribution Preferences: Evidence from an Online Experiment*

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Abstract

This paper examines how information about relative performance affects distributive preferences and beliefs about the role of effort versus luck in generating inequality. Using an online experiment, participants complete a code-recognition task and are randomly assigned to one of three feedback conditions: no feedback, told they performed *above* a randomly matched peer, or told they performed *below* a randomly matched peer. They then act as impartial spectators and distribute a fixed sum between two workers whose inequality arose from differential performance. Participants told they outperformed their peer redistribute less to the lower-earning worker and score higher on an effort-attribution belief scale (0–10) than those told they underperformed. No significant effect is found for participants told they underperformed relative to the above group. These results are consistent with self-serving attribution: learning that one outperformed a peer reinforces beliefs that inequality reflects effort, which in turn reduces redistribution demand.

Keywords: redistribution, relative performance feedback, self-serving attribution, fairness, online experiment

JEL codes: C91, D31, D63, D91

*This experiment builds on the design of **?** and is inspired by the research question of **?**.

1 Introduction

Free-market economies inherently generate unequal outcomes, and while modern democratic societies rely on redistribution to mitigate these disparities, there is persistent disagreement over which inequalities are considered legitimate (**??**). A central dimension of this debate concerns the *source* of inequality: outcomes attributed to effort are typically viewed as more deserving than those arising from luck. This intuition is formalised in the accountability framework, which models redistributive preferences as judgments about deservingness and predicts lower redistribution when inequality reflects effort rather than chance (**??**). The framework finds broad empirical support: beliefs about the determinants of income are consistently among the strongest predictors of redistribution preferences (**???**).

Yet a fundamental challenge underlies this debate: in practice, the source of inequality is often ambiguous. Individual success depends on a complex combination of hard work, innate talent, family background, and circumstance — factors that are difficult to disentangle even for those directly involved (**?**). When effort and luck cannot be cleanly separated, beliefs about which is responsible are not simply passive reflections of reality; they are also shaped by personal experience and motivated reasoning (**?**). Since these beliefs directly drive redistribution preferences — as established above — whatever shapes them ultimately shapes the demand for redistribution too.

? argue that the experience of success and failure is itself a driver of redistributive polarization through the self-serving bias: individuals who succeed at a task attribute their success to effort and favour less redistribution, while those who fail attribute failure to circumstance and shift toward more. Because success and wealth are correlated, this mechanism creates a systematic wedge between rich and poor in their perceived causes of inequality — wealthy individuals, as **?** put it, "do not only oppose redistribution because they expect to be net payers, but also because the self-serving bias systematically shifts their fairness principles." Causal identification of this channel is difficult outside a controlled setting, however: in observational data one cannot tell whether successful individuals attribute their outcomes to effort because of cognitive bias or because they genuinely worked harder. Isolating the self-serving bias therefore requires exogenous variation in comparative signals — variation that holds actual performance constant while shifting only what individuals learn about their relative standing. The question this paper asks is whether such a signal — being told one outperformed or underperformed a peer — is enough to move beliefs about the role of effort in generating inequality and, through that channel, redistribution preferences.

This paper provides experimental evidence on this mechanism. Participants complete a performance task and are randomly assigned to one of three conditions: (i) *no feedback* about relative standing; (ii) told they performed *above* a randomly matched peer (*above* group); or (iii) told they performed *below* a matched peer (*below* group). Subsequently, all participants act as impartial spectators and decide how to distribute a fixed monetary sum between two workers whose inequality arose from differential performance.

The key findings are three. First, participants told they outperformed a peer redistribute significantly less in the merit scenario: \$1.87 on average versus \$2.47 in the no-feedback group, a difference of \$0.59 that is statistically significant at the 5 percent level in OLS regressions with and without controls. Second, the same group scores approximately 0.78 points higher on an effort-attribution scale (0–10), consistent with self-serving belief updating, though this difference is only marginally significant individually ($p \approx 0.086$) while the joint test across groups is significant ($p = 0.028$, ANOVA). Third, the effect is asymmetric: participants told they underperformed do not significantly change their redistribution or belief scores relative to the no-feedback baseline. Taken together, these results suggest that positive performance signals trigger self-serving attribution that translates into reduced redistributive generosity — but that negative signals do not produce a symmetric reverse effect.

These results are consistent with the self-serving attribution account of redistribution developed by **?**: individuals who perform well relative to others update their beliefs about the role of effort in generating inequality and become less redistributive. Our design extends their framework by using *randomly assigned comparative feedback*, providing clean causal identification and isolating the relative rather than absolute performance channel. One notable difference is that **?** find SSB effects on both sides — success reduces redistribution and failure increases it — whereas our evidence is concentrated on the winning side. The absence of a significant response among participants told they underperformed suggests that the SSB in our setting is more asymmetric: learning one beat a peer is enough to trigger meritocratic updating, but learning one lost does not produce a symmetric shift toward redistribution. The paper most directly relates to **?**, from whose spectator design we adapt the merit redistribution task, and to **?**, whose focus on how performance comparisons shape distributive preferences motivates our research question. Our contribution is to provide clean experimental identification of the direction and asymmetry of this effect using randomly assigned relative feedback.

The rest of the paper is structured as follows. Section 2 reviews the related literature. Section 3 describes the experimental design. Section 4 presents the results. Section **??** concludes.

2 Related Literature

This paper contributes to three strands of literature.

Beliefs about inequality and redistribution. A foundational insight from **?** is that beliefs about social mobility and the role of effort versus luck determine preferences for redistribution in equilibrium. **?** provide empirical evidence: Americans attribute poverty more to individual failure than Europeans, which partly explains the cross-country gap in redistribution. **?** model how self-serving motivated reasoning sustains two equilibria — a "belief in a just world" and a more luck-oriented view — with different steady-state redistribution levels. Our experiment directly tests the mechanism by which an informational shock (relative performance feedback) updates beliefs and behaviour.

Impartial spectator and fairness views. **?** and **?** use spectator designs to classify subjects into fairness types (libertarian, meritocratic, egalitarian) and show that a majority of individuals prefer some redistribution away from inequality that arises from luck rather than effort. We adapt their spectator design to a merit scenario and ask whether third-party redistribution decisions are affected by information about the spectator's own relative performance — a test of whether the spectator role is truly impartial.

Self-serving attribution and social comparisons. Self-serving attribution refers to the tendency to attribute successes internally (effort) and failures externally (luck or circumstance) (**??**). In an economic context, **?** document that parties in bargaining inflate self-assessments of a fair outcome in a self-serving direction. Closer to our design, **?** show that individuals who experience unfair treatment become more supportive of redistribution. **?** show that framing one's own performance as a failure (relative to a counterfactual) shapes distributive preferences; our design extends this insight by using randomly assigned comparative feedback, allowing clean causal identification. We add to this literature by focusing on the *comparative* and *asymmetric* dimension: the same absolute performance level can produce different attribution and redistribution depending on whether one learns one stands above or below a peer, and the effect appears concentrated on the winning side.

3 Experimental Design

The experiment was conducted online using JATOS and hosted the free, ESCoP-sponsored server Mindprobe. The whole design builds on the spectator design of **?**. It involved two distinct roles: *workers* and *spectators*.

Workers ($n = 2$) were recruited through Prolific and completed the code-recognition task under real monetary incentives. They were paid a base fee of \$4 plus a performance-based bonus of up to \$8, for a maximum possible payout of \$12 each. They were informed in advance that their earnings would initially depend on their relative performance against each other—the higher-scoring worker would receive \$8 while the other would receive \$0—and that a third party (a spectator) would subsequently decide whether and how much to redistribute between them.

Spectators ($n = 134$) were recruited through personal networks (friends, family, and colleagues from two workplaces), completed the full experiment online in approximately 10 minutes, and received no monetary compensation. The experiment consisted of two parts: a code-recognition task and a redistribution decision task in which they acted as impartial spectators. Spectators were informed before making their redistribution decisions that the workers were real individuals who had completed the same task and would receive actual monetary payouts determined by one randomly selected spectator's allocation. The impartial-spectator design thus combines the removal of material self-interest (spectators have no stake in the allocation) with real stakes for the workers, ensuring that decisions carry genuine consequences and lending the design greater external validity.

3.1 Part 1: Code-Recognition Task

Participants (both workers and spectators) completed a code-recognition task over one 20 seconds round of practice and five rounds of 45 seconds each. In each round, a target code was displayed at the top of the screen. Participants were shown a grid of codes and had to check as many instances of the target as possible. Scoring: $+1$ point per correct identification, -1 point per false alarm, and no points were subtracted for a miss. The maximum possible score was 200 points; sample means range from 91 to 97 points across treatment groups (Table 1).[1]

Just after finishing, participants answered three pre-feedback self-report questions: (i) effort exerted (slider, 0–10); (ii) guessed score (numeric, range -200 to 200); (iii) relative performance prediction ("I would perform better than or the same as the other participant" vs. "I would perform worse").

3.2 Treatments: Relative Performance Feedback

After the self-report, participants were randomly assigned to one of three feedback conditions.

[1] This was the end of the study for the two workers; spectators proceeded to the next stage.

1. **No feedback.** Participants were informed they could proceed to Part 2 without seeing their score or any comparison information.
2. **Above.** Participants were told: "You performed above your randomly matched peer."
3. **Below.** Participants were told: "You performed below your randomly matched peer."

Spectators assigned to the above condition were matched to the worst-performing worker, and those assigned to the below condition were matched to the best-performing worker. This matching was done randomly. As a result, a spectator who scored well could nonetheless receive a "below" message — because their reference point was the highest scorer — and a spectator who scored poorly could receive an "above" message. Crucially, the feedback was not deceptive: each spectator genuinely did outperform or underperform their assigned reference worker. What was exogenously manipulated was the identity of that reference, not the direction of the comparison itself. Spectators were shown their actual score and informed of the matching procedure only during the debriefing phase at the end of the experiment.

3.3 Part 2: Distribution Decisions

After the (non)feedback screen, all participants acted as impartial spectators and made a redistribution decision adapted from **?**.

Redistributive Decision in a Merit Scenario

Participants read that two real participants — the two workers who had previously completed the same task — were identified as Worker A and Worker B. Worker A had received a larger initial payment because they performed better; Worker B received nothing because they performed worse. Participants were asked to decide whether and how to redistribute the given endowment between the two workers. The decision was presented as a set of discrete options covering the full range from keeping full inequality to reversing it.

After the redistribution decision, participants completed a short demographics survey (age, gender, and education) followed by the key belief measure of this study. They were asked to respond to the following question using a slider:

> *"To what extent do you think the initial income of the participants in the previous task was determined by effort versus luck?"*

The slider ranged from 0 ("Completely determined by luck") to 10 ("Completely determined by effort"). This variable (`effort_luck`) is the hypothesised mediator between relative performance feedback and redistribution preferences.

3.4 Key Outcome and Mediator Variables

- **Redistribution (merit):** USD allocated to Worker B in the merit scenario. Primary outcome variable.

- **Effort–luck belief (`effort_luck`):** Score on the 0–10 slider described above. Hypothesised mediator.

3.5 Sample

Spectators were recruited through my personal networks: friends, family, and employees at two workplaces. The analysis sample consists of $N = 146$ spectators after excluding non-engagers (zero score and zero clicks). This is a convenience sample and is therefore not nationally representative; the main implication for external validity is that findings describe the direction of the self-serving attribution effect under incentivized conditions but may not generalise to other populations in terms of magnitudes. Importantly, treatment assignment was random within this sample, so internal validity is not compromised.

Table 1 presents descriptive statistics by treatment group. Groups are balanced on pre-determined covariates: age ($p = 0.94$), gender ($p = 0.31$), task score ($p = 0.30$), and prior belief about relative performance ($p = 0.16$), confirming that randomisation was successful. The effort–luck belief score differs significantly across groups ($p = 0.043$), which is itself a key outcome; importantly, this measure is elicited *after* the feedback treatment, so it cannot confound treatment assignment.

Table 1: Descriptive Statistics by Treatment Group

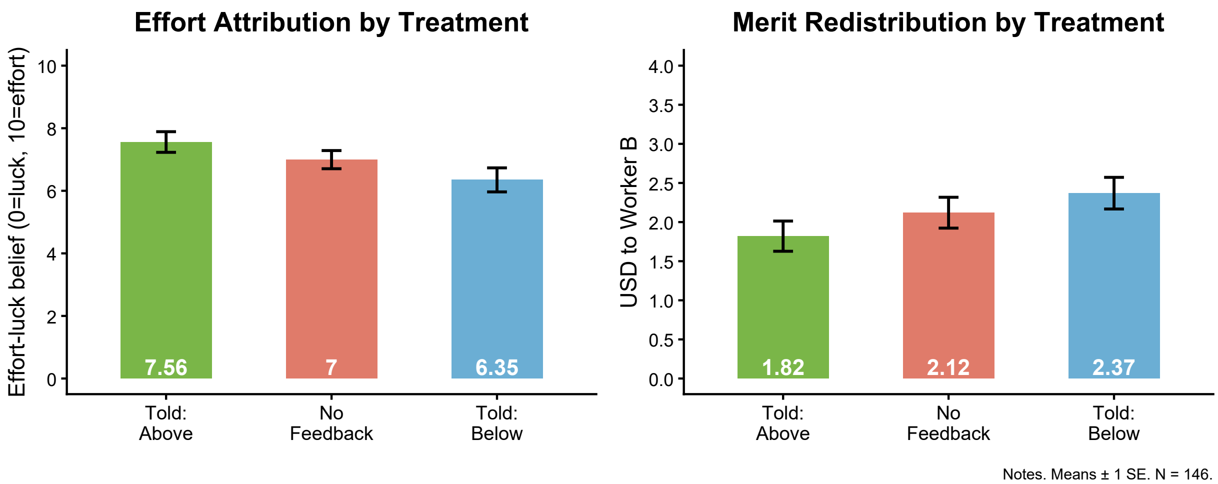
	Told: Above	No Feedback	Told: Below	p-value
Redistribution – merit ($)	1.82 (1.37)	2.12 (1.39)	2.37 (1.37)	0.151
Redistribution – luck ($)	2.62 (1.58)	3.27 (1.36)	3.10 (1.39)	0.110
Effort–luck belief (0–10)	7.56 (2.31)	7.00 (2.03)	6.35 (2.58)	0.043
Task score	97.72 (29.30)	92.50 (26.37)	88.91 (27.98)	0.300
Age	34.86 (13.58)	35.84 (14.97)	35.57 (15.12)	0.942
Female (%)	60.00 (49.49)	62.00 (49.03)	73.91 (44.40)	0.308
Predicted better (%)	–	–	–	0.160
N	50	50	46	146

Mean (SD) for continuous variables; mean × 100 (SD × 100) for binary.
p-values from one-way ANOVA (continuous) or chi-squared test (binary). Redistribution in USD.

4 Results

4.1 Effort–Luck Beliefs

Figure 1: Effort Attribution and Merit Redistribution by Treatment Group



Notes. Left panel: mean effort–luck belief slider (0 = 100% luck; 10 = 100% effort), measured after the merit redistribution decision. Right panel: mean amount allocated (in USD) to the lower-earning worker in the merit scenario. Bar heights correspond to group means; whiskers represent ± 1 standard error. $N = 146$.

Figure 1 (left panel) displays mean effort–luck belief by treatment group. Participants told they outperformed their matched peer attribute the most to effort ($\overline{EL}_{\text{above}} = 7.56$, SD = 2.31), followed by the no-feedback group ($\overline{EL}_{\text{nofb}} = 7.00$, SD = 2.03) and the below group ($\overline{EL}_{\text{below}} = 6.35$, SD = 2.58). A one-way ANOVA rejects equality of group means at the 5 percent level ($p = 0.043$).

Columns 1–2 of Table 2 present OLS regressions of `effort_luck` on treatment dummies, using the above group as the baseline. The no-feedback group scores 0.56 points lower on the effort-attribution scale (SE = 0.44, $p = 0.21$), a difference in the expected direction that falls short of conventional significance. The below group scores 1.21 points lower than the above group (SE = 0.51, $p < 0.05$), a statistically significant difference consistent with self-serving attribution. Both coefficients are essentially unchanged with demographic and task-score controls (column 2), ruling out confounding by pre-existing differences. Three participants did not complete the effort–luck slider; columns 1–2 use $N = 143$.

Table 2: Treatment Effects on Effort Attribution and Redistribution

	(1)	(2)	(3)	(4)
Constant	7.559***	6.451***	1.820***	3.513***
	(0.334)	(1.175)	(0.195)	(0.740)
No Feedback	−0.563	−0.610	0.300	0.272
	(0.444)	(0.440)	(0.279)	(0.275)
Told: Below	−1.210**	−1.283**	0.550*	0.498*
	(0.512)	(0.530)	(0.283)	(0.297)
Age		0.032**		−0.020**
		(0.015)		(0.009)
Female		0.222		−0.136
		(0.406)		(0.240)
Task score		−0.002		−0.010**
		(0.007)		(0.005)
N	143	143	146	146
R^2	0.044	0.091	0.026	0.069
Controls (age, female, score)	No	Yes	No	Yes

* p <0.1, ** p <0.05, *** p <0.01

Notes. OLS with HC3-robust SEs in parentheses. Baseline: Told Above. Cols. (1)–(2): effort–luck belief (0 = luck, 10 = effort). Cols. (3)–(4): USD redistributed to lower-earning worker (merit scenario). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.2 Redistribution in the Merit Scenario

The right panel of Figure 1 shows mean redistribution in the merit scenario. The above group allocates the least to the lower-earning worker (\$1.82, SD = \$1.37), compared with the no-feedback group (\$2.12, SD = \$1.39) and the below group (\$2.37, SD = \$1.37). The one-way ANOVA yields $p = 0.151$, not significant at the omnibus level.

Columns 3–4 of Table 2 report the main OLS regressions ($N = 146$). Relative to the above group, the no-feedback group redistributes \$0.30 more (SE = \$0.28, $p = 0.28$) and the below group redistributes \$0.55 more (SE = \$0.28, $p < 0.10$). Adding controls for age, gender, and task score leaves both estimates stable (column 4: no-feedback \$0.27, SE = \$0.28; below \$0.50, SE = \$0.30, $p < 0.10$). The pattern is consistent with self-serving attribution: participants who learn they outperformed a peer become the least redistributive, with the difference relative to the below group reaching marginal statistical significance.

4.3 Heterogeneity by Surprise

To probe the attribution mechanism, we classify participants in the relative-feedback group by whether the feedback was surprising given their prior expectation. *Positively surprised* participants predicted they would not outperform their peer but were told they did. *Negatively surprised* participants predicted they would outperform but were told they underperformed. Table **??** reports OLS regressions within the relative-feedback group, with confirmed-outcome participants (those whose feedback matched their prior prediction) as the baseline. If self-serving attribution drives the result, the redistribution effect should be sharpest among those for whom the positive signal was unexpected — unexpected confirmation of high standing is arguably more diagnostic and self-affirming than expected confirmation. Table **??** (columns 3–4) reports these results. The results are consistent with this prediction, though the relative-feedback subsample is underpowered for this analysis and caution is warranted.

5 Conclusion

This paper provides experimental evidence that relative performance feedback shapes beliefs about the role of effort in generating inequality and, through that channel, redistribution preferences. Participants told they outperformed their peer attribute significantly more of the task outcome to effort than those told they underperformed (gap of 1.21 points on a 0–10 scale, $p < 0.05$), and redistribute less in the merit scenario (gap of \$0.55 relative to the below group, $p < 0.10$). No significant redistribution effect is found relative to the no-feedback group, though the above–below gap is statistically significant and consistent with a self-serving attribution channel.

Taken together, these findings are consistent with a self-serving attribution account of redistribution preferences: learning one outperformed a peer reinforces the belief that inequality reflects effort, which in turn reduces willingness to redistribute in settings where merit is salient. That the effect is most clearly identified by contrasting participants who received positive versus negative comparative signals — rather than feedback versus no feedback — suggests the *direction* of the relative signal, not mere exposure to comparison, is the operative mechanism. This asymmetry mirrors the well-documented pattern in the psychology of self-serving attribution (**?**) and has implications for the design of performance information systems.

These findings have implications for the study of inequality and redistribution in a world where performance comparisons are increasingly salient — on professional platforms, social

Table 3: Mediation Analysis

	(1)	(2)	(3)	(4)
Constant	3.361***	4.772***	2.083***	3.268***
	(0.445)	(0.744)	(0.143)	(0.908)
No Feedback	0.171	0.144		
	(0.275)	(0.274)		
Told: Below	0.292	0.239		
	(0.278)	(0.292)		
Effort–luck belief	−0.204***	−0.200***		
	(0.052)	(0.053)		
Age		−0.012		−0.015
		(0.009)		(0.013)
Female		−0.094		0.295
		(0.233)		(0.314)
Task score		−0.010**		−0.009
		(0.004)		(0.006)
N	143	143	96	96
R^2	0.137	0.169	0.000	0.040
Controls (age, female, score)	No	Yes	No	Yes

* p <0.1, ** p <0.05, *** p <0.01

Notes. OLS with HC3-robust SEs in parentheses. Dependent variable: USD redistributed (merit scenario). Cols. (1)–(2): full sample; includes effort–luck belief to assess mediation; compare treatment coefficients with Table 2 cols. (3)–(4). Cols. (3)–(4): relative-feedback group only. "Positively surprised": told above but predicted below. "Negatively surprised": told below but predicted above. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

networks, and in educational settings. The self-serving updating documented here suggests that rising relative visibility of performance rankings could systematically reduce support for redistribution among those who "win" such comparisons, potentially contributing to the polarisation of redistribution preferences.

Future work should (i) increase sample size to improve power, particularly for the heterogeneity and mediation analyses; (ii) test whether the effect persists when participants receive absolute score information alongside relative feedback, to disentangle comparative and informational channels; and (iii) conduct a formal mediation analysis to quantify the share of the redistribution effect that operates through the effort-attribution belief measure.

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